

```
1 test_instance_1 = pd.DataFrame([{\n2     'age': 35, 'balance': 1500, 'day': 15, 'duration': 300,\n3     'campaign': 2, 'previous': 0, 'pdays': -1,\n4     'job': 'technician', 'marital': 'married', 'education': 'secondary',\n5     'default': 'no', 'housing': 'yes', 'loan': 'no',\n6     'contact': 'cellular', 'month': 'may', 'poutcome': 'unknown'\n7 }])\n8\n9 test_instance_2 = pd.DataFrame([{\n10     'age': 58, 'balance': 4200, 'day': 5, 'duration': 600,\n11     'campaign': 1, 'previous': 2, 'pdays': 180,\n12     'job': 'retired', 'marital': 'single', 'education': 'tertiary',\n13     'default': 'no', 'housing': 'no', 'loan': 'no',\n14     'contact': 'cellular', 'month': 'aug', 'poutcome': 'success'\n15 }])\n16\n17 for i, instance in enumerate([test_instance_1, test_instance_2], 1):\n18     pred = model.predict(instance)[0]\n19     prob = model.predict_proba(instance)[0][1]\n20     label = 'YES' if pred == 1 else 'NO'\n21     print(f"Instance {i}: {label} ({prob:.2%})")\n\n✓ [4] 100ms\n\nInstance 1: NO (18.73%)\nInstance 2: YES (97.63%)
```

300

-

+

no

▼

Number of contacts this campaign

2

-

+

Has housing loan?

yes

▼

Previous contacts

0

-

+

Has personal loan?

no

▼

Days since last contact (-1 if never)

-1

-

+

Contact type

cellular

▼

Last contact month

may

▼

Previous campaign outcome

unknown

▼

Predict

Prediction: NO

Probability of subscribing: 18.73%

600	-	+	no	▼
Number of contacts this campaign			Has housing loan?	
1	-	+	no	▼
Previous contacts			Has personal loan?	
2	-	+	no	▼
Days since last contact (-1 if never)			Contact type	
180	-	+	cellular	▼
			Last contact month	
			aug	▼
			Previous campaign outcome	
			success	▼

Predict

Prediction: YES

Probability of subscribing: 97.63%

